



Stem cells from extracted teeth

Researchers from the University of Nevada have developed a device to extract 80 percent of the stem cells from tooth pulp. <http://digit.in/ToothCrkr>

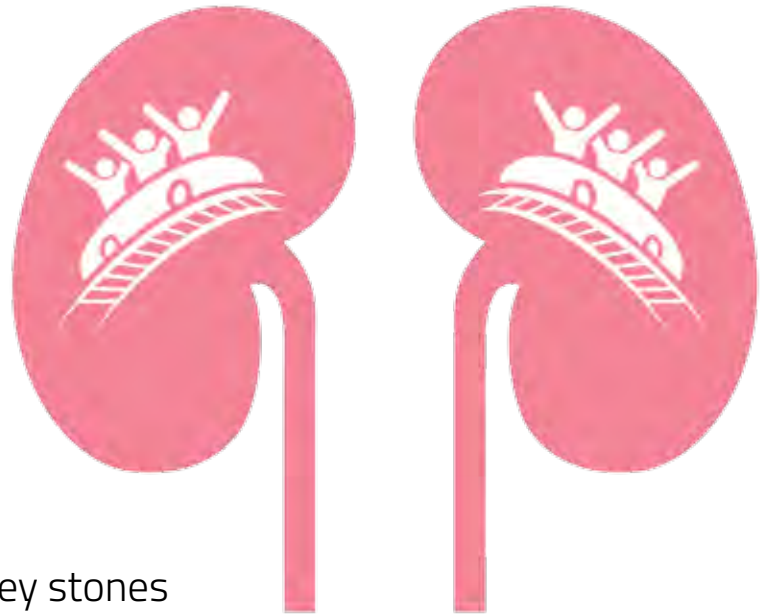


Ancient atmosphere on Moon

According to study, volcanic eruptions could have created a temporary lunar atmosphere billions of years ago <http://digit.in/MoonAir>

A roller-coaster cure

Who knew that passing kidney stones and ridding roller-coasters could be synonymous with each other?



Arnab Mukherjee | arnab@digit.in

R

iding a roller-coaster generates a lot of different emotions in people - some of us are terrified, some

exhilarated, and then there are those who don't really feel anything. But today what we are going to talk about is a fourth group of people - the ones who get cured of an ailment after a round on their favourite roller coaster.

When a group of medical researchers saw many of their patients returning from Disney's Magic Kingdom in Orlando, Florida with a previously detected kidney stone passed (for one of them, even three in a row), they decided to take matters into their own hands and test it out for themselves. Dr. David Wartinger and Marc Mitchell, from the Michigan State University College of Osteopathic Medicine in East Lansing, developed a 3D-printed silicone cast of a patient's kidney. They didn't stop with the realism at that - they filled it with kidney stones and urine and headed to the park.

Holding the model at about kidney height, about 20 rides into their experiment, the researchers reached an understanding that a roller coaster ride, especially on the Big Thunder Mountain Railroad roller coaster at the theme park, was good enough to dislodge a kidney stone and allow it to pass through the urinal tract - with as high a success rate as 64 percent at the back of the coaster.

Keep in mind that the researchers hadn't really rented out the entire ride for themselves - so they had no control of where they sat on the coaster. That, combined with the fact that almost every single one of these rides had a different stone alignment (small stone at bottom, large in middle, medium on top etc.), and one could see reasonable doubt in this study. Additionally, this method is only even applicable for people with small kidney stones. The dimensions of the stones used in the experiment were 4.5 mm³, 13.5 mm³, or 64.6 mm³ for small, medium and large respectively. Even for the largest

size, if it was placed in the upper position, there was a 100% rate of passing.

Due to several factors like low water intake, dairy and wheat prominent diet, prevalence of obesity and diabetes, people in India run a high risk of developing kidney stones during their lifetimes. In fact, we have held the world records for most number of kidney stones removed from a single patient as well as the largest kidney stone removed successfully at one point of time. In such a situation, people with small kidney stones could opt for the roller coaster treatment - which would definitely prove to be cheaper than a full-fledged extraction surgery. Although, as stated above, this may not work under certain conditions and it is always advisable to go to your urologist first and consult them about the idea. Then, if there's a go ahead, you could literally have fun while getting rid of a kidney stone. If that's not killing two birds with one stone, we don't know what is. 

Study: <http://digit.in/KidneyCoaster>